

Jupyter notebook with GPUs

From your local machine login into sherlock

```
>ssh username@sherlock.stanford.edu
```

Ask for a node with a GPU (24 GB ram and 12 cores) for 4 hours

```
>salloc -c 12 --mem=24GB -p serc --gres gpu:1 --time=04:00
```

Pay attention to the node that you are allocated. It will look something like

```
salloc: Nodes sh03-14n09 are ready for job
```

In this case you have been allocated node “sh03-14n09”

If you haven’t already set a password for your jupyter notebook

```
>jupyter notebook password
```

Load an environment that includes jupyter and any other packages you might need

```
> module load py-jupyter/1.0.0_py39
```

Start jupyter lab or a jupyter notebook. Choose a random port number between 45000 and 89000

```
>jupyter-notebook --no-browser --port=50234 --ip=`hostname -s`
```

Open another terminal on your computer

```
> ssh -f -L 50234:sh03-14n09:50234 clapp@login.sherlock.stanford.edu sleep 300
```

Where 50234 is my random port number and sh03-14n09 is the node I was allocated. Now you can open the notebook on your local machine with your random port number

<http://localhost:50234>

Important options for both notebooks and sbatch scripts

-c Number of cores - Even if your job is GPU based you will want at least 4 cores per GPU

`--mem` Amount of CPU memory. Your job will automatically die if you run out of memory. Or GPU nodes have about 125 GB per GPU so it is best to over estimate

`--p serc` Always use the serc partition, the gpu partition is for the entire university and is massively oversubscribed

`--time=days-hours:minutes:seconds` Always specify the amount of time you want a node

`--gres=gpu:X` where X is the number of GPUs

`--constraint="GPU_SKU:A100_SXM4"` Only run on the A100 (newer GPUs)